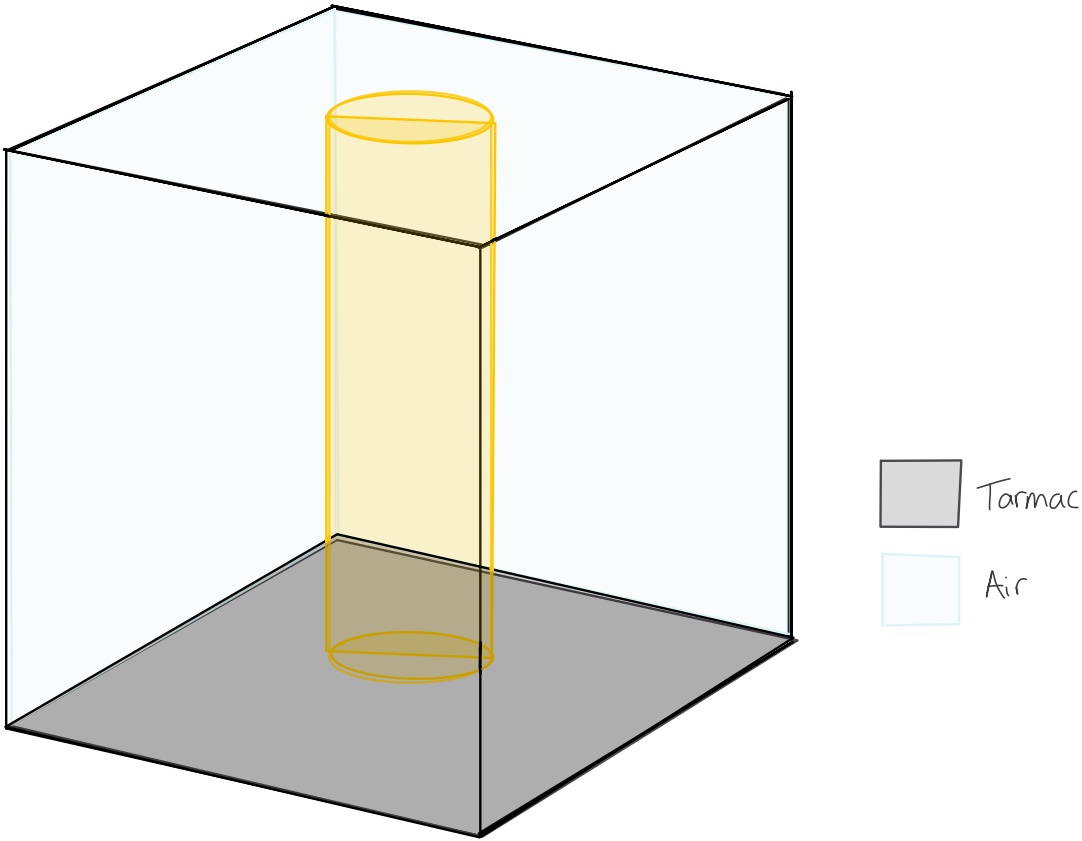
Tarmac
Air

We consider the poisson problem in 3 dimensions on the unitary cube

$$\Omega = [0,1]\times[0,1]\times[0,1]:$$

$$-\Delta u(x,y,z) = f(x,y,z), \quad (x,y,z) \in \Omega,$$

$$u(x,y,z) = g(x,y,z), \quad (x,y,z) \in \partial\Omega$$

where $\partial\Omega$ denotes the boundary of the unitary square

We define the following discretisation of the unitary square as points (x_i, y_j, z_k) for $0 \leq i,j,k \leq N+1$ such that for the step size $h = \frac{1}{N+1}$ we have $x_i = ih$, $y_j = jh$ and $z_k = kh$.

We define the following discrete laplacian as

$$\Delta_h u_{i,j,k} \simeq \frac{k}{h^2}\left(-6u_{i,j,k} + u_{i+1,j,k} + u_{i-1,j,k} + u_{i,j-1,k} + u_{i,j+1,k} + u_{i,j,k-1} + u_{i,j,k+1}\right),$$

where $u_{i,j,k} = u(x_i, y_j, z_k)$, where k is a constant representing thermal conductivity.

We will enumerate every point of the grid (x_i, y_j, z_k) from left to right, from front to back and from bottom to top (so the first point is $(0,0,0)$), then it is possible to list without ambiguity the N^3 points with the rule:

$$(u_h)_n = u_h(x_i, y_j, z_k), \quad n = (k-1)N^2 + (j-1)N + i, \quad 1 \leq i,j,k \leq N.$$

We now consider

$$f(x,y,z) = \begin{cases} 0.005 & (x-0.5)^2+(y-0.5)^2 < 0.2^2, \quad 0 < z \leq 1 \\ 0 & (x-0.5)^2+(y-0.5)^2 > 0.2^2, \quad 0 < z \leq 1 \end{cases}$$

$$g(x,y,z) = \begin{cases} 50 & (x-0.5)^2+(y-0.5)^2 < 0.2^2, \quad z = 0 \\ 15 & (x-0.5)^2+(y-0.5)^2 > 0.2^2, \quad z = 0 \\ 0 & z = 1, \quad 0 \leq x, y \leq 1 \\ 0 & x = 0, \quad 0 \leq y, z \leq 1 \\ 0 & x = 1, \quad 0 \leq y, z \leq 1 \\ 0 & y = 0, \quad 0 \leq x, z \leq 1 \\ 0 & y = 1, \quad 0 \leq x, z \leq 1 \end{cases}$$